

From Trajectories to Instructions: Language-Conditioned Meta-Reinforcement Learning

1 Introduction

This material includes the hyper-parameters and architecture details along with all the results obtained during the evaluation of the experiments performed in our paper. The following tables compare the performance of different models across different baselines, reporting the results in the form of average steps the agent requires to solve the task with the success rate. Further, an ablation study is performed on our framework to analyze the importance of language component of our approach. Each section presents the tables along with detailed descriptions.

2 Hyper-parameters and Architecture

The following section contains the hyper-parameters and architectural details used for experimentation.

Table 1: Hyperparameters used in experiments

Hyperparameter	Value
Meta-Iterations	200
Tasks per Meta-Batch	5
Trajectories per Task (K)	40
Inner Learning Rate (Task)	$1e^{-4}$
Outer Learning Rate (Meta)	$1e^{-3}$
Inner loop adaptation steps	1
Actor Policy Head (MLP)	(64, 64), Tanh
Text Encoder	all-MiniLM-L6-v2
Task Embedding Size	384
Parameter Adapter (MLP)	384 $\rightarrow$ 128 $\rightarrow$ Policy Params, Tanh

3 Comparison of LA-MAML against MAML, Language-conditioned Policy and ANIL

In this section, we present the performance comparison of LA-MAML with 3 mentioned baselines across various environments under different configurations. The results highlight the advantages of the LA-MAML framework in terms of adaptation and task performance. The reported values are the average number of steps taken by each policy, accompanied by their standard deviation. We also evaluate the performance using success rate which depicts the number of times the agent successfully completed the task in the given maximum time steps.

Key Observations:

- LA-MAML consistently outperforms all three baselines across nearly all environments and configurations.
- The performance gap is most pronounced in environments requiring sequential door interactions, such as `OpenDoor`, `OpenDoorLoc`, and `OpenDoorsOrder`, where MAML’s success rate drops as low as 0.25 to 0.35 while LA-MAML maintains rates above 0.86.

- In simpler navigation tasks such as **GoToLocal** and **GoToObjDoor**, all methods perform comparably, which shows that language conditioning provides a larger benefit in structurally complex tasks.
- The Language-Conditioned policy and ANIL baselines show intermediate performance but consistently fall behind LA-MAML, particularly as room size and the number of distractors increase. For instance, in **PickupDist** with 5 distractors, Language conditioned policy achieves only 0.35 SR compared to LA-MAML’s 0.60. This demonstrates that meta-learning alone is insufficient and that language-guided parameter adaptation is an important factor.

Table 2: Performance comparison on **GoToLocal**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	2	300	0.3	17.36 $\pm$ 13.53	0.98	20.41 $\pm$ 12.34	0.99	37.75 $\pm$ 17.96	1.00	15.73 $\pm$ 6.59	1.00
6	4	300	0.3	48.37 $\pm$ 24.00	0.88	51.26 $\pm$ 14.98	0.89	79.80 $\pm$ 31.25	0.86	40.94 $\pm$ 14.62	0.94
7	3	300	0.3	30.72 $\pm$ 15.56	0.94	40.71 $\pm$ 20.05	0.96	52.97 $\pm$ 19.37	0.97	34.37 $\pm$ 17.98	0.97
7	5	300	0.3	82.58 $\pm$ 35.90	0.81	63.94 $\pm$ 23.83	0.88	96.72 $\pm$ 38.82	0.83	90.62 $\pm$ 40.85	0.82
8	2	300	0.3	26.57 $\pm$ 13.73	0.99	32.28 $\pm$ 15.23	0.99	68.92 $\pm$ 38.36	0.93	37.23 $\pm$ 23.58	0.96
8	4	300	0.3	55.91 $\pm$ 31.39	0.89	56.55 $\pm$ 17.20	0.93	102.93 $\pm$ 22.54	0.88	65.71 $\pm$ 18.83	0.92
Overall Average				43.59 $\pm$ 22.35	0.91	44.19 $\pm$ 17.27	0.94	73.18 $\pm$ 28.05	0.91	47.43 $\pm$ 20.41	0.93

Table 3: Performance comparison on **PickupDist**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	2	500	0.3	136.97 $\pm$ 36.32	0.93	236.86 $\pm$ 68.66	0.63	342.73 $\pm$ 98.62	0.42	182.52 $\pm$ 28.79	0.83
6	4	500	0.3	212.81 $\pm$ 70.62	0.74	288.78 $\pm$ 74.03	0.51	402.53 $\pm$ 60.35	0.28	244.14 $\pm$ 57.46	0.74
7	3	500	0.3	172.32 $\pm$ 41.83	0.87	300.19 $\pm$ 75.19	0.47	342.59 $\pm$ 58.69	0.41	228.64 $\pm$ 70.13	0.78
7	5	500	0.3	259.31 $\pm$ 98.69	0.60	359.60 $\pm$ 72.30	0.36	364.28 $\pm$ 59.76	0.35	349.72 $\pm$ 61.52	0.47
8	2	500	0.3	143.77 $\pm$ 42.37	0.90	244.09 $\pm$ 89.16	0.63	320.87 $\pm$ 62.59	0.52	207.93 $\pm$ 76.39	0.80
8	4	500	0.3	240.74 $\pm$ 82.34	0.75	289.63 $\pm$ 89.33	0.55	377.35 $\pm$ 71.95	0.33	323.78 $\pm$ 65.54	0.55
Overall Average				194.15 $\pm$ 62.03	0.80	286.69 $\pm$ 78.11	0.53	358.39 $\pm$ 68.66	0.36	256.12 $\pm$ 59.97	0.70

Table 4: Performance comparison on **GoToObjDoor**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
env	1	300	0.3	28.46 $\pm$ 20.99	0.98	51.53 $\pm$ 29.70	0.98	74.95 $\pm$ 44.70	0.93	39.97 $\pm$ 30.76	0.99
env	2	300	0.3	41.30 $\pm$ 22.95	0.97	53.02 $\pm$ 29.14	0.97	72.67 $\pm$ 26.64	0.96	48.57 $\pm$ 34.44	0.96
env	3	300	0.3	57.71 $\pm$ 24.47	0.92	57.29 $\pm$ 20.87	0.97	89.05 $\pm$ 48.19	0.93	60.88 $\pm$ 28.01	0.95
env	4	300	0.3	73.40 $\pm$ 22.52	0.89	70.86 $\pm$ 31.10	0.93	97.17 $\pm$ 29.94	0.92	85.36 $\pm$ 24.93	0.93
env	5	300	0.3	73.42 $\pm$ 22.30	0.88	89.32 $\pm$ 30.31	0.88	103.89 $\pm$ 48.23	0.86	91.99 $\pm$ 33.81	0.89
Overall Average				54.86 $\pm$ 22.65	0.93	64.40 $\pm$ 28.22	0.95	87.55 $\pm$ 39.54	0.92	65.35 $\pm$ 30.39	0.94

Table 5: Performance comparison on **GoToOpen**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	4	1200	0.3	407.41 $\pm$ 107.00	0.82	504.88 $\pm$ 131.34	0.76	545.83 $\pm$ 101.94	0.76	561.51 $\pm$ 137.55	0.70
6	8	1200	0.3	444.15 $\pm$ 135.63	0.79	536.76 $\pm$ 149.02	0.72	537.25 $\pm$ 147.66	0.74	482.47 $\pm$ 111.51	0.75
7	4	1200	0.3	564.69 $\pm$ 134.87	0.66	519.24 $\pm$ 101.20	0.72	612.15 $\pm$ 137.70	0.63	578.53 $\pm$ 150.84	0.73
7	7	1200	0.3	488.66 $\pm$ 145.64	0.71	757.02 $\pm$ 163.24	0.57	624.45 $\pm$ 247.80	0.66	589.03 $\pm$ 147.64	0.69
7	10	1200	0.3	573.14 $\pm$ 135.26	0.70	606.76 $\pm$ 115.13	0.62	685.15 $\pm$ 195.48	0.59	678.04 $\pm$ 105.18	0.59
8	6	1200	0.3	738.43 $\pm$ 158.59	0.49	685.96 $\pm$ 169.65	0.58	702.44 $\pm$ 116.64	0.58	804.65 $\pm$ 173.97	0.54
8	8	1200	0.3	657.27 $\pm$ 199.64	0.59	667.28 $\pm$ 135.70	0.59	672.23 $\pm$ 127.44	0.60	658.99 $\pm$ 126.27	0.61
8	10	1200	0.3	695.42 $\pm$ 107.88	0.58	702.44 $\pm$ 123.19	0.57	742.95 $\pm$ 166.10	0.55	658.47 $\pm$ 180.51	0.57
8	12	1200	0.3	646.11 $\pm$ 223.16	0.58	762.27 $\pm$ 152.27	0.55	810.90 $\pm$ 89.46	0.47	705.58 $\pm$ 151.89	0.54
Overall Average				579.48 $\pm$ 149.74	0.66	638.07 $\pm$ 137.86	0.63	659.26 $\pm$ 147.80	0.62	635.25 $\pm$ 142.82	0.64

Table 6: Performance comparison on **OpenDoor**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	Step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	800	0.3	267.87 $\pm$ 61.87	0.76	472.41 $\pm$ 143.85	0.44	378.94 $\pm$ 81.44	0.71	506.36 $\pm$ 127.75	0.42
6	env	800	0.3	145.46 $\pm$ 72.95	0.96	622.00 $\pm$ 97.68	0.25	481.37 $\pm$ 84.17	0.58	537.00 $\pm$ 102.53	0.39
7	env	800	0.3	136.97 $\pm$ 46.33	0.98	544.23 $\pm$ 107.97	0.37	420.04 $\pm$ 129.08	0.67	586.36 $\pm$ 71.29	0.32
8	env	800	0.3	198.71 $\pm$ 90.65	0.90	621.41 $\pm$ 67.65	0.26	498.20 $\pm$ 74.30	0.52	590.33 $\pm$ 97.17	0.30
9	env	800	0.3	272.38 $\pm$ 101.44	0.86	543.62 $\pm$ 85.92	0.35	538.57 $\pm$ 82.08	0.49	585.23 $\pm$ 78.90	0.34
Overall Average				204.28 $\pm$ 74.65	0.89	560.73 $\pm$ 100.61	0.33	463.42 $\pm$ 90.21	0.59	561.06 $\pm$ 95.53	0.35

Table 7: Performance comparison on **OpenDoorLoc**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	800	0.3	94.78 $\pm$ 54.83	0.96	432.38 $\pm$ 92.89	0.49	357.50 $\pm$ 75.50	0.67	472.73 $\pm$ 89.88	0.44
6	env	800	0.3	98.82 $\pm$ 67.85	1.00	520.40 $\pm$ 106.57	0.37	358.32 $\pm$ 95.26	0.73	478.04 $\pm$ 158.96	0.43
7	env	800	0.3	105.01 $\pm$ 65.97	0.99	506.65 $\pm$ 102.71	0.41	468.31 $\pm$ 128.33	0.55	492.53 $\pm$ 112.32	0.44
8	env	800	0.3	189.94 $\pm$ 154.65	0.94	605.18 $\pm$ 131.68	0.28	469.34 $\pm$ 122.19	0.54	476.52 $\pm$ 123.62	0.47
9	env	800	0.3	189.35 $\pm$ 113.50	0.94	515.80 $\pm$ 143.20	0.42	440.82 $\pm$ 129.28	0.58	520.06 $\pm$ 102.03	0.41
Overall Average				135.58 $\pm$ 91.36	0.97	516.08 $\pm$ 115.41	0.39	418.86 $\pm$ 110.11	0.61	487.98 $\pm$ 117.36	0.44

Table 8: Performance comparison on **OpenDoorsOrder**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML		Lang-Cond		ANIL	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	1200	0.3	189.70 $\pm$ 123.68	0.76	283.51 $\pm$ 141.48	0.58	265.61 $\pm$ 112.91	0.64	224.10 $\pm$ 110.32	0.76
6	env	1200	0.3	181.57 $\pm$ 106.14	0.93	373.81 $\pm$ 168.16	0.69	396.87 $\pm$ 199.88	0.59	357.07 $\pm$ 185.26	0.69
7	env	1200	0.3	246.59 $\pm$ 158.62	0.93	507.56 $\pm$ 248.97	0.66	526.88 $\pm$ 290.47	0.66	525.83 $\pm$ 272.96	0.60
8	env	1200	0.3	434.40 $\pm$ 238.32	0.87	703.00 $\pm$ 387.79	0.62	667.05 $\pm$ 309.78	0.65	606.65 $\pm$ 304.07	0.72
9	env	1200	0.3	535.70 $\pm$ 235.11	0.93	900.77 $\pm$ 417.95	0.66	986.10 $\pm$ 425.78	0.58	860.23 $\pm$ 383.17	0.72
Overall Average				317.59 $\pm$ 172.37	0.88	553.73 $\pm$ 272.87	0.64	568.50 $\pm$ 267.76	0.62	514.78 $\pm$ 251.16	0.70

4 Comparing LA-MAML with MAML’s few-shot adaptation

In this section the results show that even training MAML for 1,2,3 step adaptation, it still underperforms LA-MAML in most of the environments. We propose 2 types of tables:

1. Table 9 shows combined results obtained for all environments as we show in main paper for all other evaluations (we could not include in the main paper as we do for other due to space constraints).
2. Table 10-16 shows all the results obtained while performing experiments across multiple configurations for each environment.

Key Observations:

- Even on increasing the number of MAML inner-loop adaptation steps it is unable to close the gap with LA-MAML in most environments.
- In `OpenDoor` and `OpenDoorLoc`, the 1-step MAML performance drops drastically (SR of 0.33 and 0.39 respectively), and while 3-step adaptation partially recovers performance (SR of 0.86 and 0.94), it does so at the cost of significantly more gradient steps during inference. However, LA-MAML achieves comparable or superior success rates with a single step adaptation.
- The environments in which MAML variants of 2-step and 3-step shows competitive behavior are `GoToLocal` and `GoToObjDoor`. These environments incorporates simpler tasks where the language signal provides less additional benefit.

Table 9: Average steps comparison across all environments. (Lower is better)

Environment	LA-MAML	MAML (1-step)	MAML (2-step)	MAML (3-step)
GoToLocal	44.04 $\pm$ 24.82	44.19 $\pm$ 17.27	40.66 $\pm$ 23.10	36.47 $\pm$ 21.22
PickupDist	179.73 $\pm$ 48.02	286.52 $\pm$ 78.11	231.73 $\pm$ 59.68	203.97 $\pm$ 48.63
GoToObjDoor	61.07 $\pm$ 27.27	64.40 $\pm$ 28.22	49.27 $\pm$ 29.26	53.73 $\pm$ 29.86
GoToOpen	582.25 $\pm$ 149.31	638.07 $\pm$ 137.86	617.67 $\pm$ 147.48	633.96 $\pm$ 146.60
OpenDoor	212.82 $\pm$ 83.39	560.73 $\pm$ 100.61	385.36 $\pm$ 96.44	233.17 $\pm$ 87.10
OpenDoorLoc	119.33 $\pm$ 74.63	516.08 $\pm$ 115.41	483.68 $\pm$ 108.41	157.26 $\pm$ 98.85
OpenDoorsOrder	271.90 $\pm$ 142.85	553.73 $\pm$ 272.87	424.57 $\pm$ 221.17	347.77 $\pm$ 185.84

Table 10: Performance comparison on `GoToLocal`. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	2	300	0.3	15.06 $\pm$ 10.92	0.98	20.41 $\pm$ 12.34	0.99	19.76 $\pm$ 15.92	0.98	21.34 $\pm$ 12.19	0.97
6	4	300	0.3	57.50 $\pm$ 19.56	0.86	51.26 $\pm$ 14.98	0.89	58.24 $\pm$ 25.56	0.87	43.22 $\pm$ 31.02	0.90
7	3	300	0.3	41.76 $\pm$ 25.55	0.92	40.71 $\pm$ 20.05	0.96	30.96 $\pm$ 18.21	0.95	43.04 $\pm$ 31.94	0.92
7	5	300	0.3	62.80 $\pm$ 46.66	0.86	63.94 $\pm$ 23.83	0.88	62.86 $\pm$ 39.05	0.89	46.09 $\pm$ 25.39	0.91
8	2	300	0.3	40.09 $\pm$ 19.80	0.94	32.28 $\pm$ 15.23	0.99	41.78 $\pm$ 19.30	0.95	33.54 $\pm$ 13.66	0.96
8	4	300	0.3	47.00 $\pm$ 26.44	0.92	56.55 $\pm$ 17.20	0.93	30.34 $\pm$ 20.55	1.00	31.59 $\pm$ 13.14	0.98
Overall Average				44.04 $\pm$ 24.82	0.91	44.19 $\pm$ 17.27	0.94	40.66 $\pm$ 23.10	0.94	36.47 $\pm$ 21.22	0.94

Table 11: Performance comparison on **PickupDist**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	2	500	0.3	138.53 $\pm$ 58.63	0.92	236.86 $\pm$ 68.66	0.63	153.71 $\pm$ 41.77	0.91	143.83 $\pm$ 51.59	0.87
6	4	500	0.3	200.57 $\pm$ 58.35	0.78	288.78 $\pm$ 74.03	0.51	296.10 $\pm$ 85.76	0.57	242.83 $\pm$ 71.94	0.68
7	3	500	0.3	161.34 $\pm$ 44.42	0.87	300.19 $\pm$ 75.19	0.47	260.64 $\pm$ 55.10	0.66	196.32 $\pm$ 60.68	0.77
7	5	500	0.3	276.15 $\pm$ 55.46	0.64	359.60 $\pm$ 72.30	0.36	333.50 $\pm$ 59.51	0.46	375.07 $\pm$ 49.09	0.38
8	2	500	0.3	146.44 $\pm$ 22.89	0.94	244.09 $\pm$ 89.16	0.63	155.29 $\pm$ 50.54	0.88	121.86 $\pm$ 22.12	0.97
8	4	500	0.3	155.37 $\pm$ 48.39	0.91	289.63 $\pm$ 89.33	0.55	191.15 $\pm$ 65.40	0.81	143.92 $\pm$ 36.36	0.93
Overall Average				179.73 $\pm$ 48.02	0.84	286.52 $\pm$ 78.11	0.53	231.73 $\pm$ 59.68	0.71	203.97 $\pm$ 48.63	0.77

Table 12: Performance comparison on **GoToObjDoor**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
env	1	300	0.3	26.93 $\pm$ 16.06	0.99	51.53 $\pm$ 29.70	0.98	35.80 $\pm$ 33.02	0.98	31.90 $\pm$ 27.80	0.98
env	2	300	0.3	39.48 $\pm$ 22.75	0.97	53.02 $\pm$ 29.14	0.97	38.71 $\pm$ 30.25	0.99	42.54 $\pm$ 34.22	0.97
env	3	300	0.3	75.57 $\pm$ 24.22	0.91	57.29 $\pm$ 20.87	0.97	46.54 $\pm$ 20.84	0.99	45.70 $\pm$ 30.77	0.96
env	4	300	0.3	68.94 $\pm$ 24.70	0.89	70.86 $\pm$ 31.10	0.93	58.30 $\pm$ 37.16	0.95	65.82 $\pm$ 15.72	0.91
env	5	300	0.3	94.45 $\pm$ 48.61	0.83	89.32 $\pm$ 30.31	0.88	67.02 $\pm$ 25.03	0.94	82.67 $\pm$ 40.80	0.89
Overall Average				61.07 $\pm$ 27.27	0.92	64.40 $\pm$ 28.22	0.95	49.27 $\pm$ 29.26	0.97	53.73 $\pm$ 29.86	0.94

Table 13: Performance comparison on **GoToOpen**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
6	4	1200	0.3	370.56 $\pm$ 158.96	0.88	504.88 $\pm$ 131.34	0.76	466.31 $\pm$ 91.91	0.77	425.70 $\pm$ 99.57	0.79
6	8	1200	0.3	440.15 $\pm$ 141.83	0.78	536.76 $\pm$ 149.02	0.72	468.68 $\pm$ 166.11	0.81	483.38 $\pm$ 141.73	0.74
7	4	1200	0.3	551.75 $\pm$ 139.32	0.67	519.24 $\pm$ 101.20	0.72	584.00 $\pm$ 117.56	0.66	666.05 $\pm$ 128.88	0.58
7	7	1200	0.3	532.55 $\pm$ 111.44	0.72	757.02 $\pm$ 163.24	0.57	683.87 $\pm$ 142.11	0.59	556.43 $\pm$ 217.10	0.67
7	10	1200	0.3	690.49 $\pm$ 146.03	0.55	606.76 $\pm$ 115.13	0.62	627.59 $\pm$ 176.05	0.63	714.16 $\pm$ 122.70	0.57
8	6	1200	0.3	670.41 $\pm$ 152.95	0.61	685.96 $\pm$ 169.65	0.58	644.99 $\pm$ 149.63	0.58	735.68 $\pm$ 92.63	0.48
8	8	1200	0.3	746.84 $\pm$ 138.47	0.52	667.28 $\pm$ 135.70	0.59	677.71 $\pm$ 135.37	0.57	661.56 $\pm$ 169.78	0.57
8	10	1200	0.3	553.47 $\pm$ 193.94	0.66	702.44 $\pm$ 123.19	0.57	690.87 $\pm$ 192.56	0.55	702.74 $\pm$ 213.01	0.53
8	12	1200	0.3	684.00 $\pm$ 160.82	0.57	762.27 $\pm$ 152.27	0.55	715.05 $\pm$ 156.01	0.55	759.97 $\pm$ 134.03	0.51
Overall Average				582.25 $\pm$ 149.31	0.66	638.07 $\pm$ 137.86	0.63	617.67 $\pm$ 147.48	0.63	633.96 $\pm$ 146.60	0.60

Table 14: Performance comparison on **OpenDoor**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	800	0.3	267.46 $\pm$ 156.86	0.74	472.41 $\pm$ 143.85	0.44	307.06 $\pm$ 104.02	0.75	230.61 $\pm$ 124.38	0.89
6	env	800	0.3	169.55 $\pm$ 83.66	0.91	622.00 $\pm$ 97.68	0.25	384.72 $\pm$ 89.36	0.66	145.03 $\pm$ 80.47	0.93
7	env	800	0.3	144.00 $\pm$ 55.34	0.96	544.23 $\pm$ 107.97	0.37	339.93 $\pm$ 132.99	0.70	202.41 $\pm$ 52.64	0.87
8	env	800	0.3	203.90 $\pm$ 53.27	0.94	621.41 $\pm$ 67.65	0.26	429.36 $\pm$ 80.71	0.61	301.99 $\pm$ 107.29	0.82
9	env	800	0.3	279.21 $\pm$ 67.80	0.84	543.62 $\pm$ 85.92	0.35	465.75 $\pm$ 75.12	0.59	285.79 $\pm$ 70.73	0.81
Overall Average				212.82 $\pm$ 83.39	0.88	560.73 $\pm$ 100.61	0.33	385.36 $\pm$ 96.44	0.66	233.17 $\pm$ 87.10	0.86

Table 15: Performance comparison on **OpenDoorLoc**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	800	0.3	115.65 $\pm$ 81.85	0.93	432.38 $\pm$ 92.89	0.49	450.92 $\pm$ 103.64	0.46	146.07 $\pm$ 85.98	0.88
6	env	800	0.3	90.64 $\pm$ 60.82	1.00	520.40 $\pm$ 106.57	0.37	429.78 $\pm$ 108.40	0.49	138.75 $\pm$ 66.09	0.96
7	env	800	0.3	100.19 $\pm$ 69.30	1.00	506.65 $\pm$ 102.71	0.41	539.60 $\pm$ 101.96	0.35	136.01 $\pm$ 87.49	0.96
8	env	800	0.3	130.42 $\pm$ 66.32	0.96	605.18 $\pm$ 131.68	0.28	504.04 $\pm$ 140.79	0.43	156.66 $\pm$ 108.52	0.96
9	env	800	0.3	159.75 $\pm$ 94.88	0.97	515.80 $\pm$ 143.20	0.42	494.04 $\pm$ 87.27	0.43	208.79 $\pm$ 146.18	0.92
Overall Average				119.33 $\pm$ 74.63	0.97	516.08 $\pm$ 115.41	0.39	483.68 $\pm$ 108.41	0.43	157.26 $\pm$ 98.85	0.94

Table 16: Performance comparison on **OpenDoorsOrder**. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance							
Room	Dist	Horizon	step-size	LA-MAML		MAML (1-step)		MAML (2-step)		MAML (3-step)	
				Steps	SR	Steps	SR	Steps	SR	Steps	SR
5	env	1200	0.3	203.19 $\pm$ 132.83	0.75	283.51 $\pm$ 141.48	0.58	221.58 $\pm$ 117.53	0.78	184.48 $\pm$ 100.54	0.82
6	env	1200	0.3	153.61 $\pm$ 82.31	0.96	373.81 $\pm$ 168.16	0.69	301.90 $\pm$ 220.20	0.74	261.44 $\pm$ 125.51	0.86
7	env	1200	0.3	209.70 $\pm$ 146.31	0.95	507.56 $\pm$ 248.97	0.66	424.70 $\pm$ 203.60	0.77	370.27 $\pm$ 207.93	0.78
8	env	1200	0.3	359.84 $\pm$ 148.59	0.90	703.00 $\pm$ 387.79	0.62	562.19 $\pm$ 313.71	0.71	441.88 $\pm$ 277.00	0.80
9	env	1200	0.3	433.14 $\pm$ 204.23	0.86	900.77 $\pm$ 417.95	0.66	612.46 $\pm$ 250.79	0.68	480.78 $\pm$ 218.20	0.87
Overall Average				271.90 $\pm$ 142.85	0.88	553.73 $\pm$ 272.87	0.64	424.57 $\pm$ 221.17	0.74	347.77 $\pm$ 185.84	0.83

5 Ablation study to analyze the importance of language during Inference

This section illustrates the results obtained when only policy parameters are used during inference, while the model was originally trained on both global and language parameters. It provides insights into the trade-offs of using only the policy parameters for inference in comparison to using both global and language parameters.

Key Observations:

- Removing the language component at inference time leads to a consistent degradation in performance across all environments, which shows that the language module plays an important role during task execution.
- The effect is especially severe in **OpenDoor** and **OpenDoorLoc**, where removing language causes the average steps to more than double ($193.90 \rightarrow 450.68$ and $141.93 \rightarrow 435.60$ respectively), while success rates drop by over 0.35.
- However, in simpler environments such as **GoToLocal**, the SR marginally improves to 0.94 despite the absence of language, likely because the task complexity is low enough to be solved through policy parameters alone, but at the cost of increased average steps.

Table 17: Ablation study on GoToLocal: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
6	2	300	0.3	17.83 ± 11.87	0.98	38.29 ± 17.05	0.99
6	4	300	0.3	31.59 ± 18.42	0.95	52.56 ± 19.67	0.95
7	3	300	0.3	25.78 ± 13.27	0.96	64.30 ± 24.58	0.92
7	5	300	0.3	86.55 ± 42.75	0.77	64.90 ± 33.53	0.92
8	2	300	0.3	31.95 ± 20.42	0.97	63.57 ± 20.29	0.95
8	4	300	0.3	72.21 ± 52.22	0.84	71.97 ± 29.18	0.91
Overall Average				44.32 ± 26.49	0.91	59.27 ± 24.05	0.94

Table 18: Ablation study on PickupDist: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
6	2	500	0.3	125.46 ± 44.83	0.88	190.98 ± 52.13	0.74
6	4	500	0.3	234.30 ± 55.31	0.73	279.09 ± 30.69	0.57
7	3	500	0.3	188.15 ± 40.38	0.85	274.47 ± 47.13	0.62
7	5	500	0.3	300.19 ± 65.34	0.58	319.90 ± 66.05	0.48
8	2	500	0.3	162.49 ± 49.40	0.85	233.89 ± 38.79	0.75
8	4	500	0.3	204.31 ± 74.66	0.76	284.93 ± 60.19	0.61
Overall Average				202.48 ± 54.99	0.78	263.88 ± 49.16	0.63

Table 19: Ablation study on GoToObjDoor: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
env	1	300	0.3	30.85 ± 24.73	0.99	79.05 ± 42.94	0.92
env	2	300	0.3	37.07 ± 16.97	0.98	82.14 ± 35.06	0.92
env	3	300	0.3	51.02 ± 21.17	0.93	76.63 ± 48.42	0.90
env	4	300	0.3	89.63 ± 39.03	0.84	95.42 ± 37.61	0.87
env	5	300	0.3	90.61 ± 42.38	0.83	83.68 ± 40.05	0.89
Overall Average				59.84 ± 28.86	0.91	83.38 ± 40.82	0.90

Table 20: Ablation study on `GoToOpen`: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
6	4	1200	0.3	328.19 $\pm$ 89.47	0.91	506.37 $\pm$ 158.85	0.72
6	8	1200	0.3	421.18 $\pm$ 102.30	0.82	549.06 $\pm$ 228.18	0.70
7	4	1200	0.3	479.60 $\pm$ 134.89	0.78	630.46 $\pm$ 199.55	0.59
7	7	1200	0.3	549.42 $\pm$ 139.33	0.70	564.49 $\pm$ 126.94	0.66
7	10	1200	0.3	640.50 $\pm$ 103.88	0.63	745.97 $\pm$ 109.71	0.49
8	6	1200	0.3	634.68 $\pm$ 145.37	0.65	738.39 $\pm$ 144.80	0.48
8	8	1200	0.3	712.96 $\pm$ 166.61	0.55	748.18 $\pm$ 213.57	0.49
8	10	1200	0.3	711.74 $\pm$ 151.50	0.57	695.12 $\pm$ 176.12	0.54
8	12	1200	0.3	706.72 $\pm$ 155.93	0.55	746.20 $\pm$ 214.90	0.50
Overall Average				576.11 $\pm$ 132.14	0.68	658.25 $\pm$ 174.74	0.57

Table 21: Ablation study on `OpenDoor`: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
5	env	800	0.3	223.45 $\pm$ 62.68	0.82	346.04 $\pm$ 75.56	0.69
6	env	800	0.3	166.21 $\pm$ 35.85	0.93	443.29 $\pm$ 84.38	0.57
7	env	800	0.3	113.68 $\pm$ 44.95	0.99	478.57 $\pm$ 134.76	0.51
8	env	800	0.3	220.04 $\pm$ 64.52	0.94	479.34 $\pm$ 90.61	0.56
9	env	800	0.3	246.13 $\pm$ 93.77	0.92	506.18 $\pm$ 120.08	0.53
Overall Average				193.90 $\pm$ 60.35	0.92	450.68 $\pm$ 101.08	0.57

Table 22: Ablation study on `OpenDoorLoc`: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
5	env	800	0.3	156.06 $\pm$ 93.46	0.89	362.07 $\pm$ 161.77	0.68
6	env	800	0.3	80.18 $\pm$ 49.41	1.00	359.35 $\pm$ 120.44	0.66
7	env	800	0.3	99.53 $\pm$ 71.27	0.98	415.31 $\pm$ 158.59	0.59
8	env	800	0.3	172.97 $\pm$ 121.33	0.95	485.96 $\pm$ 111.67	0.48
9	env	800	0.3	200.93 $\pm$ 146.17	0.88	555.31 $\pm$ 112.93	0.40
Overall Average				141.93 $\pm$ 96.33	0.94	435.60 $\pm$ 133.08	0.56

Table 23: Ablation study on **OpenDoorsOrder**: comparison of LA-MAML against inference using only global policy parameters (θ) at test time. Reported values are average steps $\pm$ standard deviation and success rate (SR). (Lower is better)

Testing Setup				Performance			
Room	Dist	Horizon	step-size	LA-MAML		Ablation During Inference	
				Steps	SR	Steps	SR
5	env	1200	0.3	186.07 $\pm$ 130.87	0.79	315.15 $\pm$ 124.11	0.51
6	env	1200	0.3	166.60 $\pm$ 84.72	0.98	394.32 $\pm$ 196.02	0.59
7	env	1200	0.3	221.42 $\pm$ 148.65	0.94	539.89 $\pm$ 241.37	0.58
8	env	1200	0.3	390.16 $\pm$ 203.97	0.89	790.85 $\pm$ 343.36	0.51
9	env	1200	0.3	533.37 $\pm$ 331.68	0.89	1063.25 $\pm$ 386.48	0.44
Overall Average				299.52 $\pm$ 179.98	0.90	620.69 $\pm$ 258.27	0.53